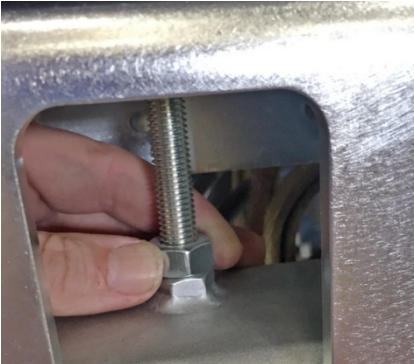
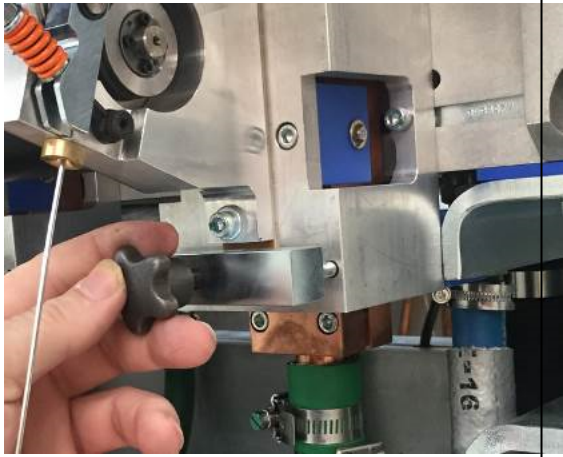


Maintenance Requirement	Equipment & Task to Be Performed
Annually	Check the location fasteners for the evaporation source assembly
Figure 12.2-142 Example of a technician checking the tightness of the locating bolts	
Figure 12.2-143 Example of a technician checking the tightness of the locating nuts	
Figure 12.2-144 Example of a technician checking the tightness of the locating bolts	

Maintenance Requirement	Equipment & Task to Be Performed
Annually	Check for the full movement of the viewing flap. If tight, replace the internal bearings
Figure 12.2-145 Example of a technician checking the movement of the viewing flap valve	

Maintenance Requirement	Equipment & Task to Be Performed
Annually	Check the tightness of the aluminium wire feed spool assembly screws
Figure 12.2-146 Example of a technician checking the tightness of the aluminium wire feed spool carriers	

Maintenance Requirement	Equipment & Task to Be Performed
2 Years	Replace springs in live post blocks – BML Field Engineer
Figure 12.2-147 Example of a technician removing the compression springs	

Maintenance Requirement	Equipment & Task to Be Performed
5 Years	Replace all insulators– BML Field Engineer

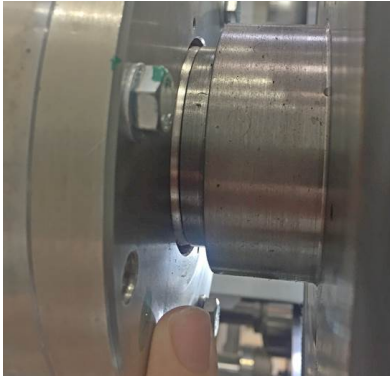
12.2.23 Substrate Winding Mechanism

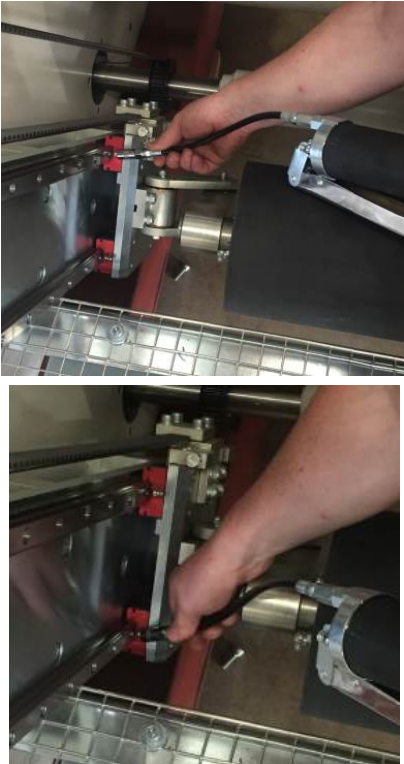
Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check all metal rollers for surface damage repair as required. Note where depressions exist in the roller surface, high spots generally exist, which can damage the substrate being processed, and need to be addressed. Deep groove damage for example will need to have the area “re-chromed” to ensure that no damage occurs due to a non-heat transfer situation.
Figure 12.2-148 Example of a technician checking the process drum surface	
Figure 12.2-149 Example of a technician checking the DTR1(WCR) surface	
Weekly	Check the area where the gas wedge is located that there is no debris trapped in between it and the process drum.

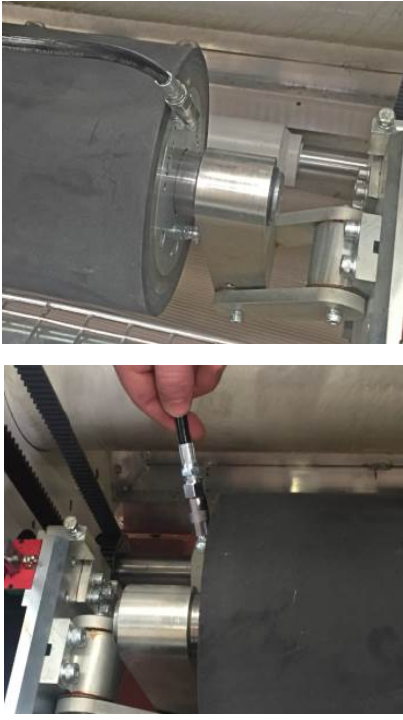
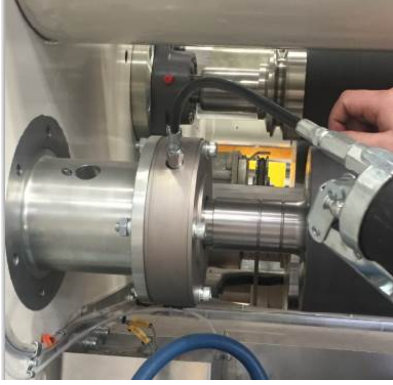
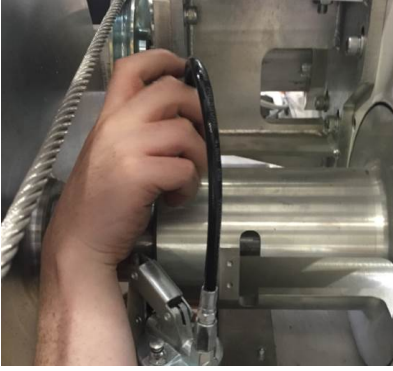
Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check all cork covered rollers for surface damage, caused by rips, knife damage, smoothness. Replace the cork covering if required.
Figure 12.2-150 Example of a technician checking a cork covered roller	
Weekly	Check all rubber rollers for physical damage. Repair if possible, or replace. Check for “tackiness”, as this can be a sign of use of incorrect cleaning fluids, attacking the roller material. Check that the “fingers” still move when pressed upon, as this allows the spreading action. If these become ridged, the spreading action will decrease, leading to wrinkle formation.
Figure 12.2-151 Example of a technician checking the unwind spreader rubber roller	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check all fibre carbon non-driven idler rollers by hand for bearing wear and stiffness, by rotating them by hand. If they are not free running, check the mounting bearings or for any mechanical binding. Replace worn parts as required. Check the surface for “chip” and any other physical damage. These should be repaired, using a suitable repair gel.
Figure 12.2-152 Example of a technician checking a carbon fibre idler roller	

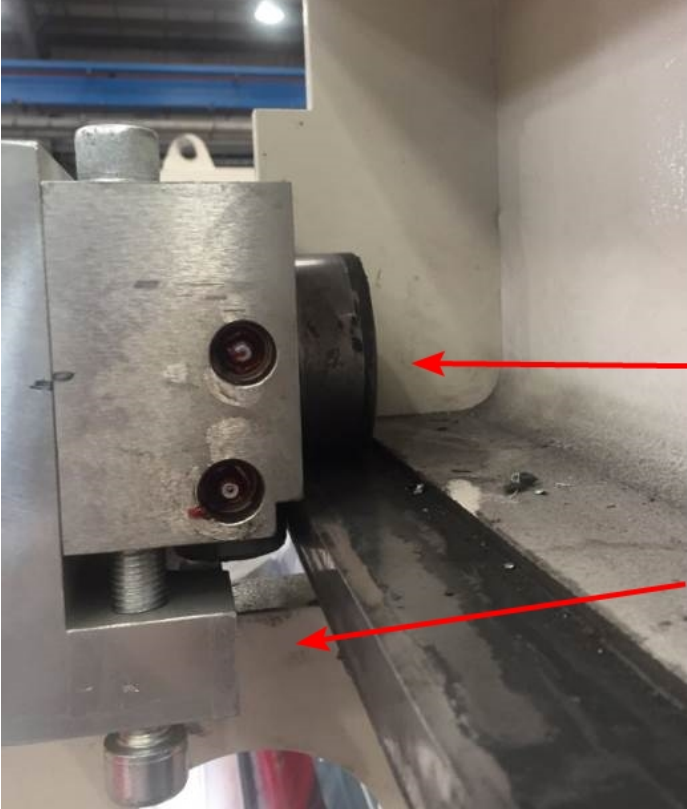
Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the layarm assembly slide rails for debris & wear. Clean & repair as required. Check that the movement is smooth in both directions using the local HMI control interface panel.
Figure 12.2-153 Example of a technician checking the unwind layarm rail assembly	
Figure 12.2-154 Example of a technician checking the rewind layarm assembly	
Weekly	Check the layarm assembly positioning belts for indications of tooth/belt wear. Check that the drive pulleys are fully engaged, and the retaining sides are in place.
Figure 12.2-155 Example of a technician checking the layarm assembly positioning belts	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check all rollers for indication of oil passing across the vacuum seal. If there is a small amount of oil present, it is normal, as this allows the seal to be lubricated. Large amounts indicate a seal failure
Figure 12.2-156 Example of a technician checking for indications of oil at the seal location on a driven roller	
Weekly	Check all fibre carbon non-driven idler rollers by hand for bearing wear and stiffness, by rotating them by hand. If they are not free running, check the mounting bearings or for any mechanical binding. Replace worn parts as required. Check the surface for “chip” and any other physical damage. These should be repaired, using a suitable repair gel.
Figure 12.2-157 Example of a technician checking the layarm idler roller	
Weekly	Visually check driven rollers for indication of oil leaks. Note that if these leaks become more severe, the vacuum performance of the machine will be affected.
Figure 12.2-158 Example of a technician greasing the layarm slide linear bearings	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Grease the layarm linear bearings on the unwind and rewind assemblies.
Figure 12.2-159 Example of a technician greasing the layarm slide linear bearings	
Weekly	Grease the unwind spreader roller at both ends as shown
Figure 12.2-160 Example of a technician greasing the unwind spreader roller bearings	

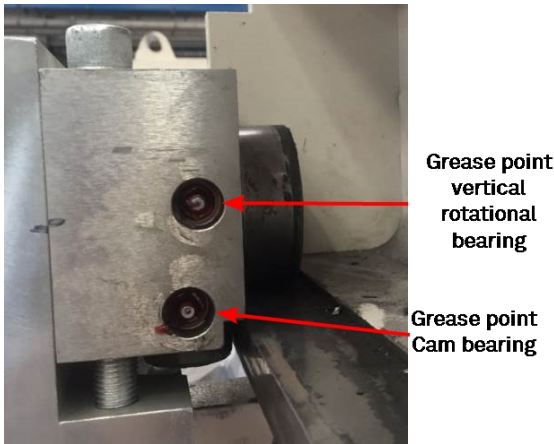
Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Grease the unwind layarm roller at both ends as shown
Figure 12.2-161 Example of a technician greasing the unwind layarm roller bearings	
Weekly	Grease the load cell roller transducer bearings at both of the roller ends as shown
Figure 12.2-162 Example of a technician greasing the load cell roller bearings	
Weekly	Grease the process drum non-driven end bearing located at the faceplate
Figure 12.2-163 Example of a technician greasing the non-driven side bearing of the process drum	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Grease the DTR2 non-driven end bearing located at the faceplate
Figure 12.2-164 Example of a technician greasing the non-driven side bearing of the DTR2 roller	
Weekly	Grease the DTR1 roller non-driven end bearing located at the faceplate
Figure 12.2-165 Example of a technician greasing the non-driven side bearing of the DTR1 roller	
Weekly	Grease the draw roller non-driven end bearing located at the faceplate
Figure 12.2-166 Example of a technician greasing the non-driven side bearing of the draw roller	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Visual inspect the upper castors (front and back), and internal chamber rails for indication of wear as highlighted
Figure 12.2-167 Upper castor assembly	 Load carrying bearing Alignment "Cam" bearings

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Grease the Kickert spreader roller rotational bearings
Figure 12.2-168 Example of a technician greasing the Kickert spreader roller apex rotational bearing	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Grease the layarm cross drive shaft, on the non-drive side of the outboard side plate
Figure 12.2-169 Example of a technician greasing the rewind layarm cross drive shaft	
Figure 12.2-170 Example of a technician greasing the unwind layarm cross drive shaft	

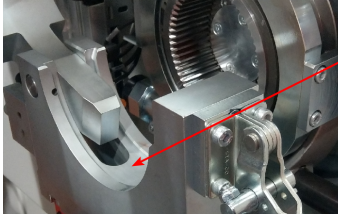
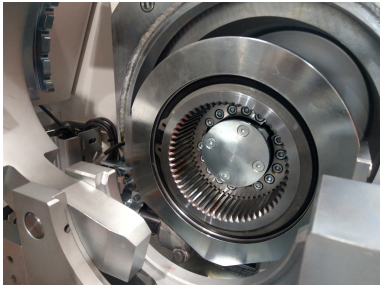
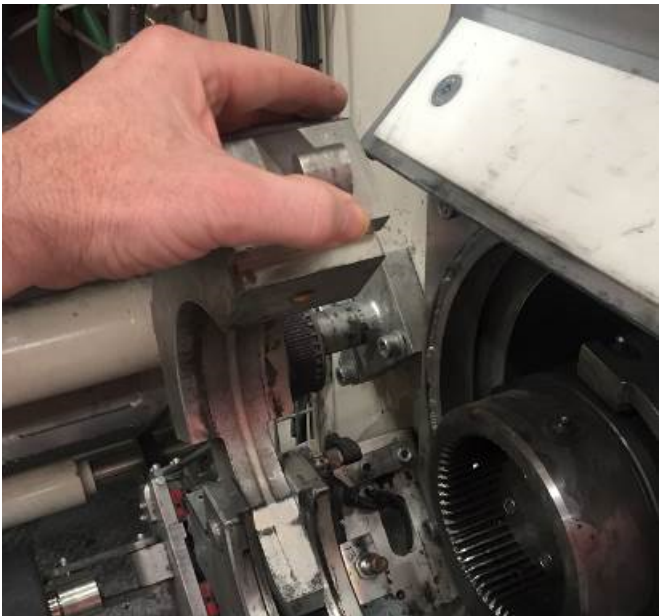
Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Grease the upper castor load bearing and guide bearings
Figure 12.2-171 Example of a technician greasing the upper rail castors	
Figure 12.2-172 The castor block grease points	
Quarterly	Check the load cell calibration as per the instructions given in this manual.
Figure 12.2-173 Example of a technician checking the load cell amplifiers	

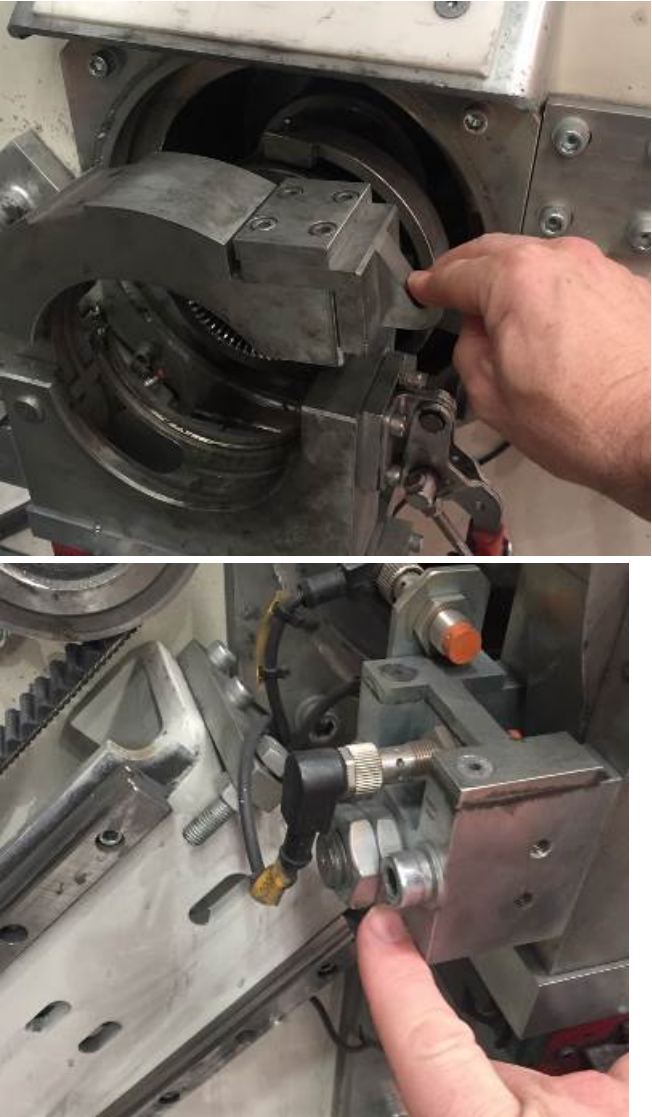
Maintenance Requirement	Equipment & Task to Be Performed
Bi Annually	Check final rewind layarm roller alignment is parallel to reel bar – BML Field Engineer

Maintenance Requirement	Equipment & Task to Be Performed
5 Years	Replace lip seals on all leadthroughs
5Years	Replace bearings on all rollers
5Years	Send 'Bow type' spreader roller back to manufacturer for roller overhaul.

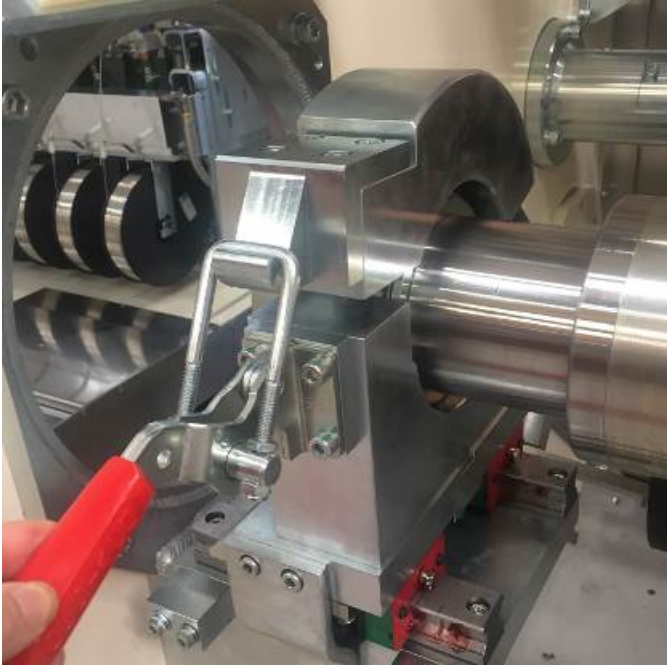
12.2.24 Chucking System & Reel Shafts

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check that the swing over clamps are free to move. If not free
Daily	Check that the chuck closed limit switches are operational, if installed

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check for wear on the chuck support surfaces
Figure 12.2-174 Example of a technician checking for wear on the chuck support surface	
Weekly	Check the rotating chucks are secure and show no signs of wear.
Figure 12.2-175 Example of a technician checking for wear on the chuck wear plates	
Weekly	Check that the swing over clamps are free to move. If they are not easy to move, investigate why and repair
Figure 12.2-176 Example of a technician checking the movement of the swing over clamp	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check that the swing over clamp, requires pressure to close it fully. This checks that the holding off spring is operational.
Figure 12.2-177 Example of a technician checking the movement of the swing over clamp, for the “holding off spring”	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check that the clamp locking toggle clamp is not damaged and when engaged, the clamp applies pressure onto the bearing.

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2-178 Example of a technician checking the movement of the toggle clamp	 A close-up photograph showing a person's hand using a pair of red-handled pliers to manipulate a metal toggle clamp mechanism. The mechanism is part of a larger industrial assembly, possibly a mold or a machine component, featuring various metal plates, bolts, and a cylindrical part. The technician is pulling on the toggle arm to test its range of motion.